

William Bachrach

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Executive Summary

- Ten years of hands-on experience in Data Science with a foundation in statistics and ML from a top STEM PhD program
- Deep experience in creating, maintaining, and leveraging large datasets for predictive and interpretive analytics
- Proven track record of translating business problems into data questions and finding actionable solutions
- Strong communication skills with a talent for explaining technical concepts to diverse audiences

Professional Experience

Owner and Senior Data Scientist

Denver, CO

Prancing Dog Motorcars LLC

November 2023 –Nov 2024

- Started a small data-driven car dealership and built bespoke R based tools to derive insight for business decisions
- Scraped primary web sources and used R-Shiny to create a simple-to-use interactive pricing model
- Leveraged Large Language Model (LLM) APIs to determine if car ads were “runners” or project cars
- Created a tool that identifies underpriced Craigslist listings and emails stakeholders twice a day using R

Lead Data Scientist

Denver, CO

Yes! Communities

February 2024 –July 2024

- Served as head data scientist at a high-growth real estate firm focusing on predictive analytics, modeling, and reporting
- Developed, implemented, & automated web scraping pipelines using R to extract critical industry data daily
- Updated predictive models to be 7% more accurate and transitioned to dynamic data housed in a SQL Server database
- Built machine learning models incorporating automated dynamic data to increase accuracy in rental pricing
- Led cross-functional collaboration with sales, engineering, and data teams to improve predictive modeling

Data Scientist Consultant

Denver, CO

Bachrach Analytics

July 2023 –February 2024

- Provided on-demand data science consulting, addressing complex data challenges
- Collaborated with NAI Shames Makovsky brokers to identify TDR certificates from over 100k city documents using OCR

Director of Data Science

New York, New York (*remote*)

Hodges Ward Elliott, Inc.

October 2020 –July 2023

- Mentored and led a team of junior data scientists, data analysts, and freelancers in a fast-paced R-centric environment
- Built a comprehensive dataset of apartment listings by web scraping using R, enabling granular market analysis
- Created “black box” machine learning models using XGBoost to identify properties that would outperform the market
- Regularly accessed multiple APIs to deliver timely information and compelling visualizations via GGPlot and R-Shiny
- Conducted extensive data mining, including text data extraction and analysis from city tax documents

Data Scientist

New York, New York

Hodges Ward Elliott, Inc.

October 2016 – October 2020

- Scraped, processed, and analyzed 41 million tax document PDFs using AWS EC2 server and S3
- Developed and deployed machine learning models to forecast NYC multifamily sale prices using novel mined data
- Extracted and analyzed NYC financial reporting metadata to assess deed transfer velocity (62 million rows)
- Managed and analyzed raw longitudinal housing survey data (1968-2017) obtained via FOIA without a data dictionary
- Integrated multiple data sources to analyze time series characteristics of NYC housing stock from 1980 to present

Staff Data Scientist

New York, New York

Messina Quantitative Research

October 2014 – May 2016

- Founding member of the data science team at a voter research predictive analytics startup (political consultancy)
- Developed and implemented novel voter targeting algorithms using SQL-housed credit bureau, public source, and survey data
- Conducted post-election ROI and causal inference analyses for cross-team, client, and executive audiences
- Designed and implemented a new "social desirability bias wrapper" methodology that significantly reduced survey bias
- Developed and analyzed A/B tests to determine the most effective messaging strategies for voters

Ph.D. Student

Tallahassee, Florida

Florida State University Department of Political Science

August 2012 – August 2014

- Served as instructor and taught highly technical material to class of 60 non-technical undergraduate students
- Led team on original research analyzing impact of foreign public opinion on state support for U.S. foreign policy
- Pilot study on method for using Google search trends to proxy traditional public opinion polls

Entrepreneurial Experience

Managing Director

Denver, CO

Prancing Dog CRE, LLC

May 2018 – Feb 2024

- Purchased 90% stake in 13 unit workforce cash flow opportunity in Commerce City, CO followed by refinance of old debt, thus recouping a significant portion of equity invested
- Acquired 5% stake in 42 unit multifamily investment in Lakewood, CO

General Partner and Data Strategist

West Palm Beach, Florida

Floyd I. Bachrach Family Limited Partnership

January 2014 – May 2018

- Responsible for sourcing and analyzing public data to guide investment strategies
- Saved a turbulent family business by leading the transition to a new business model
- Turned \$400k worth of assets into \$6.75m with 18% cash on cash returns in 5 years using 1031 exchanges

Education

- Florida State University
Master of Science in Political Science May 2014
GPA: 3.65
- Florida State University
Bachelor of Science in Political Science August 2010
GPA: 3.96 - President's list

Skills

Machine Learning / Statistics: Generative AI, Supervised, Unsupervised, Predictive Modeling, PQ/OQ, Model Deployment

Programming: R, Python, SQL, Snowflake, AWS, Linux, SAS, Stata, Docker, Bash

Tools: R-Shiny, GGPlot, Tidyverse, Tableau, Excel, MS Office, PowerPoint, Seaborn, Jira, Scikit-Learn, XGBoost, Scrapy

Data: APIs, Web Scraping, Data wrangling, Visualization, Munging, A/B Testing, Big Data Concepts

Soft Skills: Cross-Team Communication, Leadership, Mentoring, Teamwork, Business Acumen